Remington Jarrett

(810) 875-3814 | rjarrett@umich.edu | www.linkedin.com/in/remingtonjarrett | https://github.com/remyj4556

Education

University of Michigan — Ann Arbor, MI
Bachelor of Science in Computer Science - 3.85 GPA — *Expected May 2027*

Experience

Delivery Driver — June 2023 – Present
Cottage Inn Gourmet Pizza — *Fenton, MI*

- Accurately delivered customers' orders in an efficient manner.
- Assisted in the upkeep of the restaurant by sweeping, cleaning windows, and washing dishes.
- Answered phones and took in-store orders.

Projects

Strata: Voxel Engine | *C++/OpenGL*

- Engineered a multithreaded C++ voxel engine from scratch, rendered using the OpenGL graphics API, capable of maintaining stable **70+ FPS** with over **6,000** concurrently loaded chunks.
- Developed a custom task-based thread pool system, used to enable concurrent chunk generation, meshing, and rendering. Worker threads deplete a queue of callable "tasks" and sleep when complete, as to not waste CPU cycles polling an inactive thread.
- Engine architecture designed around an asynchronous chunk streaming system. During their lifetime, terrain chunks move through an atomic state machine, ensuring data correctness across multiple worker threads.
- Renderer utilizes a custom binary greedy-meshing algorithm to reduce average vertex count by over **75% (∼150)** and reduce average meshing time by over **50% (5.5ms)**.
- Reduced GPU memory usage by over **80%**, utilizing a bit-packed vertex format which stores the minimum necessary data into a single 32-bit integer. Voxel texture and material data are accessed entirely on the GPU, avoiding slow CPU-side computation and heavy data storage during the meshing sequence.

Stock Market Order Book Simulation | *C++*

- Designed a stock order matching engine, simulating real-world stock exchange logic.
- Implemented priority queue system to efficiently handle and match incoming buy/sell requests with existing orders in worst-case $O(\log(n))$ time.
- Engine tracks and continually updates median prices of each stock in the market, as well as outputting details of each successfully matched trade.
- Included an optional "time traveler" mode, using a custom algorithm to track the best possible trade across all stocks throughout the day.

Console-Based Text Editor | *C++*

- Developed a command-line text editor in C++ using a doubly linked list to efficiently manage and manipulate text in the form of a buffer.
- Implemented core text operations including insertions, deletions, and cursor movement in real time.
- Optimized the editor for memory management and performance to handle dynamic updates and large text sizes.

Technical Skills

Languages: C/C++, Python, JavaScript
Frameworks: OpenGL, Matplotlib
Developer Tools: Git, Linux, GDB, Valgrind, CMake, Perf

Relevant Courses

- Computer Architecture
- Foundations of Computer Science
- Data Structures and Algorithms
- Discrete Mathematics
- C++ Programming